

More Than the Risk: Frontier LLM Behaviour in AI Development Races Depends on the Rival

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ABSTRACT

Large language models are increasingly used to advise and act on consequential decisions. As such systems become more involved in organisational decision making, an AI-safety question arises: how would frontier agents behave when competition rewards speed while caution reduces catastrophic risk? We study this question in an idealised AI-development race, where agents repeatedly choose between slower safe development and faster unsafe development.

Across the tested frontier routes, risk responses share a qualitative direction but differ in shape. Agents respond to increasing risk, with profiles ranging from gradual adaptation to near-deterministic switching. Behaviour is also strongly opponent-dependent. In a scripted-rival study of five frontier-model endpoints that passed a frozen task-validity screen, changes in the rival’s behaviour move the agents substantially, with a magnitude that varies by route and can overlap the effect of changing the stated risk itself. Thus, self-play outcomes can reflect an interaction between two copies of a policy rather than an intrinsic risk preference.

Frontier agents also show a distinctive relationship to the theoretical and human benchmarks. An analysis extending the evolutionary comparison across the admitted panel places the four graded profiles closer in response shape to the near-neutral regime than to its strong-selection prediction, while their action trajectories generally occupy a more concentrated observed trajectory space than those of human participants: 13 of 15 admitted cells fall below every draw of a dyad-matched human null. These results suggest that AI-mediated competitive decisions may be model-specific, opponent-dependent and behaviourally concentrated, motivating evaluation at the level of interactions and trajectories rather than aggregate action rates alone. Claims are scoped to the tested routes, one prompt version and one idealised game.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

1 INTRODUCTION

LLMs are increasingly studied as agents for decision support and strategic interaction [2, 17]. As organisations consider delegating more strategic tasks to such systems, an AI-safety question follows: if a decision about how fast and how carefully to develop a powerful technology were delegated to a frontier model, what would the model decide, and would its decision look like the ones people make? The question is prospective throughout, and we do not claim that any organisation today hands a safety decision to a language model. We claim only that the behaviour a model would bring to such a decision is measurable now, in a controlled game. Controlled evaluation before widespread deployment can expose interaction effects that would be difficult to isolate in operational settings.

Competition over advanced artificial intelligence can create a safety dilemma: each company may benefit from moving faster, but safety work slows it down when rivals do not follow the same standard [5, 6, 10]. An *AI development race* is a simple game that represents this tension, in which every player chooses each round between slower Safe development and faster Unsafe development [11, 16]. The game is idealised and does not model a real AI company in full. Its value is control: it lets us hold the stated danger, the rival’s behaviour and the relative progress fixed one at a time, which no observation of real firms allows. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multiplayer version based on Han et al. [16]. We call one hosted model endpoint at one fixed configuration a *route*, because the same model name can be served under settings we do not control.

Human experiments show why an average is the wrong thing to compare. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference; exploratory analyses instead linked later Unsafe choices to the opponent’s previous action, to whether the player was ahead or behind, and to the player’s first action, so an early choice changes the later path of the interaction. A useful human and LLM comparison must therefore examine round-by-round trajectories over the first five rounds, rather than only the overall Unsafe rate.

Two problems make that comparison harder than it looks. The first is that a correctly formatted action is not an informed one. LLMs already play games of this kind as simulated participants, decision-making agents and game-playing systems [1, 2, 4], yet a model can return a valid action without tracking the state, can predict an opponent’s pattern without acting on it [2], and can distort one simulated experiment while reproducing three others [1]; in a repeated game a single such action changes the next state and every later round. Neither a plausible trajectory nor a verbal strategy report is therefore evidence here that the agent tracked the game, and an aggregate UNSAFE rate close to the human one is not evidence that it reproduces human decision dynamics or holds a general safety preference [7].

The second is that prompt-based evaluation is sensitive to presentation choices that do not change what the task asks. Formatting, option order, response-token identity, output-format requests, whitespace and persona all shift model output [24, 26, 27, 29, 33], and strategic choice is no exception: reversing action labels or re-assigning their payoffs changes choice frequencies [17], decisions move with topic, actor and world framing [25], and performance falls when a familiar default mapping is replaced by a stated alternative [30]. We therefore test wording, answer order, arithmetic disclosure, narrative framing and opaque response codes separately, and do not treat one familiar presentation as a neutral measuring instrument.

Reproducible game-based frameworks attack the same difficulty from the other side. FAIRGAME standardises repeated-game experiments across models, languages and persona conditions against game-theoretic predictions [8]; later work adds payoff scaling, a Public Goods Game and supervised recognition of canonical strategies, and finds cooperation moving opposite to its evolutionary benchmark [18, 19]. What those designs hold constant is the opponent, because the opponent in them is another copy of the model under test.

One more result shapes what a delegate population should be compared against. Simulated populations recover broad contrasts while showing less heterogeneity than the people they stand in for, in markets, cognitive phenotyping, survey populations and persona-based simulation [9, 12, 13, 15, 28], which is why position papers call social simulation defensible for exploratory use until validated against the population of interest [3]. Strategic-game evidence is similarly unsettled, with policies that shift with parameters and framing, heuristics applied more rigidly than people apply them, cooperation in place of the predicted equilibrium, and action sequences that resist reproduction from prompts alone [21, 22, 31, 32]. A set of delegates that agrees with itself more than people do is a different object from a delegate that is simply more or less cautious, so our human comparison reports within-population diversity and not only aggregate UNSAFE rates.

We therefore ask three questions about the delegate’s behaviour. *Risk*: how do frontier agents trade speed against safety as the stated chance of catastrophe rises? *Rivalry*: are those choices a property of the model, or do they emerge from the opponent the model happens to meet? *Humans and theory*: do the agents reproduce the strategic structure that evolutionary game theory predicts, or the one human participants display?

Each answer changes how the next should be read. The four graded routes respond gradually to the stated danger, while one route replaces that trade-off with a switch from always fast to always careful. But where we put the agent against scripted code instead of a second copy of itself, the rival’s stance shifts play on every route; its magnitude is checkpoint-specific and can overlap the effect of changing the stated danger. Neither external reference shows that shape: the evolutionary model predicts a switch where the agents deliver a gradient, and on raw first-five-round trajectories human participants use nearly the whole observed trajectory space while admitted routes generally use a small part of it.

Three contributions follow. First, a scripted-rival design for repeated-game evaluation and the critique it licenses. That an LLM’s self-play behaviour need not predict its behaviour against a different agent is known [20, 22], and fixed opponents are standard practice in this literature [23]; what we add is the version that matters for a safety reading, that self-play can hide the response to the treatment the study is about, measured against an undisclosed rival whose policy is fixed by code. Second, a population-level comparison with human participants read on first-five-round trajectories rather than on aggregate rates, which is what makes the compression of observed trajectories visible at all. Third, an auditable race mechanism in which the engine and not generated text resolves every round, entered through a comprehension screen that scores answers to frozen questions against ground truth and never reads a race, so that the set of routes interpreted strategically was fixed by

a rule that cannot see the behaviour being interpreted. Exploratory multiplayer results remain in the supplementary material, because the available position analysis covers two checkpoints and cannot identify a general group-size effect.

Every behavioural conclusion is scoped to its tested route, prompt version, decoding contract, provider path and run period. The labels SAFE and UNSAFE denote actions inside the experimental game and are not measurements of general LLM safety, of a subjective risk preference, or of suitability for autonomous deployment, and nothing measured here is a harm, a real safety outcome, or an action taken by an organisation.

2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1), then its generalisation to $N > 2$ companies (§2.2). Figure 1 summarises the two-player instrument and the three empirical lenses used in the main results.

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t , both simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other’s choice, and only then does the round resolve (Figure 1a). Resolution has two parts. First, progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$ with $\sigma(S) = 1.0$ and $\sigma(U) = 1.5$, so Unsafe always closes the race faster. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \quad (1)$$

Unsafe strictly dominates Safe at the stage level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon, so neither can condition on the current round being the last (Figure 1a). Termination resolves in two further steps. First, the company with the higher cumulative progress P_i^T wins a prize $B = 100$, split 50/50 on an exact tie. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

where $n_i^U(T)$ counts company i ’s Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company’s entire race payoff, accumulated stage payoffs plus the prize, is zeroed. A company that ends behind is never subject to the draw: it keeps its stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

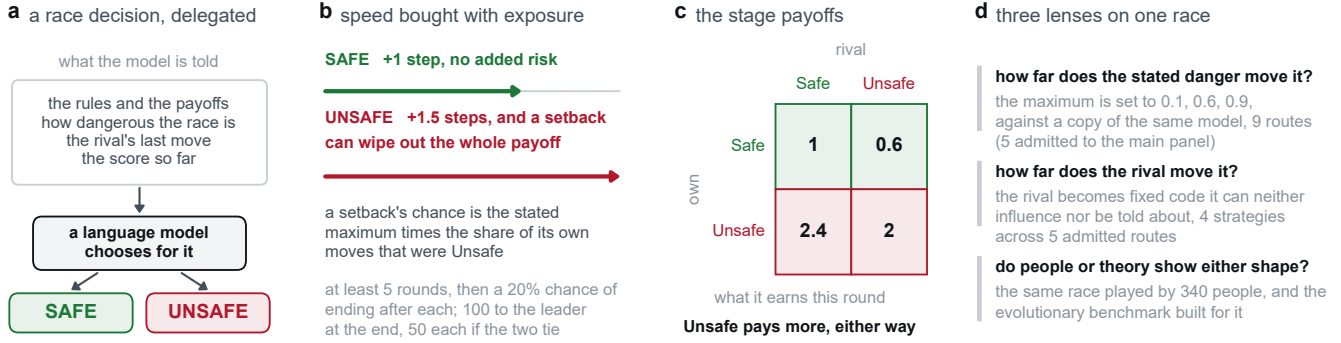


Figure 1: The game and the questions it lets us ask. (a) The model receives the rules, stated danger, rival’s last move and current score, then returns either **SAFE** or **UNSAFE**. **(b)** **SAFE** advances one step without added risk; **UNSAFE** advances faster but raises the chance that the winner’s payoff is wiped out. **(c)** The two-player stage-payoff matrix makes **UNSAFE** attractive against either rival move, so the delayed terminal risk supplies the safety tension. **(d)** The main results examine stated-risk response, scripted-rival response, and comparison with human trajectories and the evolutionary benchmark.

2.2 N -player game

The N -player game changes exactly one thing. Simultaneous sealed actions, the progress increments, the hidden minimum-five-round horizon, the terminal prize $B = 100$, and the risk formula of Equation 2 all carry over unchanged, with the prize split evenly among the companies sharing the highest progress. What cannot carry over is the stage-payoff matrix, because with more than two companies there is no single opponent to index it against. Equation 1 therefore generalises to the group-count rule of Han et al. [16], in which both payoffs depend on how many of the N companies chose Safe and not on which ones did (rather than on their identities). The engine supports $N \in \{2, 3, 4, 5\}$. Since the multiplayer runs are an exploratory extension, the rule, its parameters and its results are in the supplementary material.

2.3 Game-theoretic benchmark

The two incentives oppose each other. Stage-level dominance rewards Unsafe immediately, while across the full race Unsafe choices raise the setback risk for a winner or tied winner, so the best action can depend on the risk condition, earlier actions, and the expected response of other players.

We use four simple strategies as a theory benchmark: Always Safe (AS), Always Unsafe (AU), Conditional Safe (CS), which starts Safe and then copies the opponent, and Conditional Unsafe (CAS), which starts Unsafe and then copies the opponent. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS at 0.60, and to CS at 0.90: the theory predicts not one fixed amount of Unsafe play but a risk-dependent shift between unconditional speed, an aggressive opening and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output, because prompted self-play trajectories are not samples from an evolutionary population.

3 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

3.1 Agent protocol and units of analysis

At each decision an agent receives a versioned prompt carrying the round, the assigned maximum private risk, the disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response comes from the same pre-action snapshot, before the engine resolves actions, progress, payoffs and risk, so the environment rather than generated text determines all transitions. Raw response, parsed action, retry and parser status, pre-turn state, terminal record, configuration and seed allocation are retained for every decision.

Sampling parameters here are provider-visible contracts, not intended settings. The run manifests record a requested temperature of 0.7 that the hosted SDK forwarded on no baseline route, and no provider confirmed an effective value. Seed handling was route-specific: the SDK stripped the requested seed on the Google routes and forwarded it, application unconfirmed, on the OpenAI and Anthropic ones.

Both seats of a race are the same endpoint, so every gameplay rate below is mirror-match play, until §4.2 fixes the rival instead.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit, because decisions from one race depend on earlier states and are not separate random samples; uncertainty is grouped by race or matched repetition wherever possible. A *parse failure* is a response that does not follow the required action format, and because the fallback action changes later states, one marks the whole race as contaminated.

3.2 Evidence strata and admission

We separate confirmatory gameplay from diagnostic presentation checks. A *diagnostic pilot* is an exploratory test for possible effects or failures; it does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected,

with its protocol, route, prompt, configuration, race count and exclusion rules recorded in the run manifest. We never pool the two strata.

A paired behavioural diagnostic compares the canonical prompt against a deterministic four-row decision card that discloses the focal agent’s immediate payoff, progress and private-risk consequence for each action profile, while revealing neither the opponent’s move nor the stopping horizon. It runs on the one route here whose weights are published, screened by its own battery rather than by the screen of §3.3, so it is read only as a within-route contrast between two presentations.

Three further manipulations are described in full in the supplementary material: a persona framing with a length-matched neutral placebo, eight narrative skins that retell the same game as different stories, and a substitution of the opaque response codes *P* and *Q* for *SAFE* and *UNSAFE*. The last two are read three ways: a paired first-round comparison at an identical state, a *fixed-state replay* that asks for one action at a previously recorded state without letting the answer change later states, and a *live trajectory* in which each answer does change the next state. A task-validity screen gates which routes enter the behavioural comparison. It is an inclusion rule, not a behavioural outcome, and all gameplay results below are analysed without ranking routes by its scores.

3.3 Quality-control screen on the routes

An action that parses is not evidence that the agent followed the race. Before behavioural analysis, each route therefore answered the same frozen bank of 20 probes over six task-validity domains, three times. A route entered the study only if it cleared the fixed thresholds of 80% overall accuracy, 75% state reconstruction and 75% terminal scoring. Five of nine routes passed; the other four still played the same confirmatory baseline and remain visible in the trajectory-diversity comparison below, because two refused routes are the only ones here that reach the human reference range.

The screen reads probes and never reads a race, so gameplay cannot change its verdict. Two admitted routes pass one screening domain at the minimum attainable score, and on this nine-route roster the admitted/refused split coincides with endpoint tier naming. We therefore use the screen only as a task-validity inclusion rule, never as a capability axis or a behavioural contrast. The published-weight route used in the arithmetic diagnostic was assessed with a different battery and is not pooled with this screen. Full screening results and robustness checks are retained in the supplementary material.

4 RESULTS

These agents respond to the danger they are told about, and their behaviour is especially sensitive to the rival. Four routes have profiles closer in response shape to the near-neutral evolutionary benchmark; Claude Opus 5 is the step-like exception. All admitted routes use a more concentrated set of trajectories than people do. The four subsections take those results in turn. A route the quality-control screen of §3.3 refused appears below only where it is named as refused. Three- to five-player races, full model tables, predictive diagnostics and robustness checks are in the supplementary material.

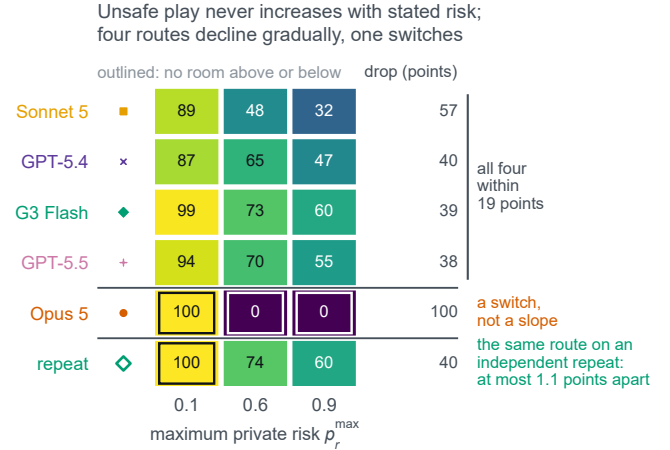


Figure 2: UNSAFE play is non-increasing with stated risk. The four graded routes fall within a 19-point band; Claude Opus 5 switches from the ceiling to the floor. The outlined cells are at a boundary and therefore leave no room for a larger or smaller observed rate. The repeat row is the same Gemini 3 Flash route collected under an independent administration.

4.1 How agents trade speed against safety

There is a shared directional response across the frontier routes. UNSAFE play is non-increasing with the stated maximum private risk on every admitted route, while the shape varies: four of the five give ground gradually and the fifth switches (Figure 2). Each route supplied 30 races and 558 dependent within-race decisions under one frozen protocol, with zero parse failures and ten races per risk cell. Risk contrasts use matched repetition blocks, and race-weighted reanalysis preserves the non-increasing response on every admitted route (supplementary material).

The four graded routes, spanning three providers, have risk-response drops within a 19-point band, from 38.2 to 57.0 percentage points. We report that band rather than an average over all five, because averaging a switch with four slopes returns 54.8 points, a figure no graded route is near and which one row produces.

That fifth route follows a threshold-like policy. Claude Opus 5 plays UNSAFE on all 186 decisions at risk 0.1 and SAFE on all 372 decisions above it, with no race inside a cell departing from that. Its 100-point figure reaches the largest value on the scale, and because no risk level was run between 0.1 and 0.6, the location of its transition is not measured.

At the highest risk the game offers, where a winner’s whole payoff is zeroed with probability up to 0.9 in proportion to how much UNSAFE it played, the four graded routes still play UNSAFE on roughly a third to three fifths of their decisions against a copy of themselves, in the $r=0.9$ column of Figure 2. This variation shows that risk-aware behaviour remains contingent on route and interaction context. Two routes can also reach a similar rate through very different sets of races, which §4.4 measures directly.

For the repeated Gemini 3 Flash route, the profile was stable across two administrations. Repeating the whole protocol on that route under the same frozen contract moved no risk cell by more

than 1.1 points over 186 decisions, against a 93.0-point spread in risk response across the nine routes, and moved its own risk response from 38.7 to 40.3. This bounds run-to-run variation for that route at this sample size; it licenses no such bound for a route collected once, which matters here because no provider confirmed applying the requested sampling seed.

Representation-equivalent prompt changes also move rates. The directional risk response survives a fully crossed narrative-by-symbol rerun on two routes, although swapping the opaque action-code mapping shifts Unsafe play by 8.3–9.5 percentage points (see the supplementary material).

4.2 The rival substantially reshapes play

In the five routes tested against scripted rivals, the observed behaviour does not support a context-independent risk profile. Across these routes, changing who is on the other side moves play substantially, but the magnitude varies by route and can overlap the effect of changing how dangerous the race is told to be. Every rate reported so far is mirror-match play, in which the rival is the route. One campaign fixes the rival instead.

The rival in that campaign is code, not a model. Five admitted routes each played the four benchmark strategies of §2.3 at all three risk levels: 60 cells, ten races each, 600 races and 5,580 route decisions with no parse failure anywhere. It is never disclosed to the route, and every one of its 5,580 recorded rival moves was recomputed afterwards from the route’s own recorded history, which no cell failed. The route’s seat is counterbalanced five and five inside each cell, and each contrast is taken inside a repetition, so the hidden horizon is differenced out. Because rival-policy assignment is fixed and exogenous within matched blocks, the direction of a difference between rival policies is identified. Conditional policies can still respond mechanically to the route’s previous move, as specified by the fixed policy.

Coverage is complete across the five routes, but decoding contracts are not homogeneous across endpoints. Cross-route differences in effect magnitude are therefore descriptive rather than controlled model comparisons; the full contract provenance is reported in the supplement.

Across these five routes, changing the scripted rival produces a positive 25.3–87.0 percentage-point shift in Unsafe play across the fifteen route-by-risk cells. Holding the rival at Always Safe, changing the stated risk from 0.1 to 0.9 produces 10.4–69.9 points, depending on the route. Both contrasts use the same ten matched blocks, so the absolute comparison is the load-bearing result, but the two effect families overlap once the conditional rivals are included. In particular, the smallest rival-stance lower bound is +16.5 points, while the largest matched-arm risk contrast is +69.9 points on Claude Opus 5. The contrast is therefore a context-sensitive descriptive comparison, not a universal ordering of rival effects over risk.

Answering the rival is general across these five endpoints. UNSAFE play rises from the rival that always plays SAFE, through the two conditional rivals, to the rival that never plays SAFE, in fourteen of fifteen route-by-risk cells, with strict ordering in twelve. The one reversal is GPT-5.5 at risk 0.1, where Conditional Unsafe reaches 100.0% and Always Unsafe is 97.8%, so this is a boundary

case rather than a clean counterexample (Figure 3a). The paired difference between the two unconditional rivals is positive in every cell. All fifteen paired-block 95% intervals have positive lower bounds, with the smallest at +16.5 points (Figure 3b). The response to the stated risk survives the change of design as well: against a fixed safe rival, no route’s UNSAFE rate rises as the stated risk rises.

The magnitude of the rival response varies markedly across checkpoints under their observed endpoint contracts: the stance contrast is largest for Claude Opus 5 at risk 0.6 (+87.0 points) and smallest for GPT-5.5 at risk 0.1 (+25.3 points); Gemini 3 Flash remains near +70 points across risks, whereas GPT-5.4 remains near +50. The level differs as well as the swing: against a rival that never plays UNSAFE, GPT-5.4 still does so 36.6%, 32.3% and 25.8% of the time. These are separate endpoint samples, so intervals that fail to overlap are a conservative sign of a difference rather than a test of one.

A mirror-match rate is therefore an interaction outcome of one policy meeting itself, rather than a stand-alone measurement of how a route treats risk. The fixed-safe and self-play designs differ in thirteen of fifteen cells; Claude Opus 5 reverses the direction at risks 0.6 and 0.9, where both rates are at or near the floor. On Gemini 3 Flash at the lowest risk, the distance is a lower bound because nine of its ten mirror-match races have every decision UNSAFE. These results motivate treating mirror-match rates as interaction outcomes rather than route-level risk preferences.

The bound on the finding is the design itself. Four reduced strategies are four points in a space of policies rather than a sample of opponents, so this establishes what a route does against *these* rivals, on one game and one prompt version, and five commercial endpoints are not a sample from a population of models.

4.3 The same incentive, a different shape

The agents and the theoretical population move in the same direction under the same incentive, but their response shapes differ because of an assumption inside the theory. At the reference setting of Domingos and Han [14], $\beta = 2$ and $\mu = \beta/Z$ predict UNSAFE play of 99.2%, 98.0% and 1.9% at risks 0.1, 0.6 and 0.9. The direct confirmatory comparison is a separate two-route block: Gemini 3 Flash gives 98.9%, 73.1% and 60.2%, and Claude Sonnet 5 gives 89.2%, 48.4% and 32.3%, over 30 races each. The other three profiles are an analysis-only extension; GPT-5.4 and GPT-5.5 decline smoothly, while Claude Opus 5 switches. Thus the difference is one of shape, not level: the $\beta = 2$ benchmark loses 98 points inside a risk step of 0.005 (Figure 4); the supplement reports the full fit.

The switch is a property of strong selection, not of the game. We swept selection strength across $\beta \in [0.001, 10]$ crossed with three mutation rules, 90 stationary-distribution cells, each estimated from independent chains with the between-chain spread retained so that a badly mixed cell cannot be mistaken for a result. As β falls towards neutrality the predicted high-risk UNSAFE rate rises off the floor and the model’s own risk response becomes graded, which is the shape the routes show. At $\beta = 0.01$ with $\mu = 0.05$ the model predicts 87.3%, 63.9% and 38.0%, the closest cell in the entire sweep to both routes, within 10.3 percentage points of Claude Sonnet 5 and 15.1 of Gemini 3 Flash in root-mean-square error.

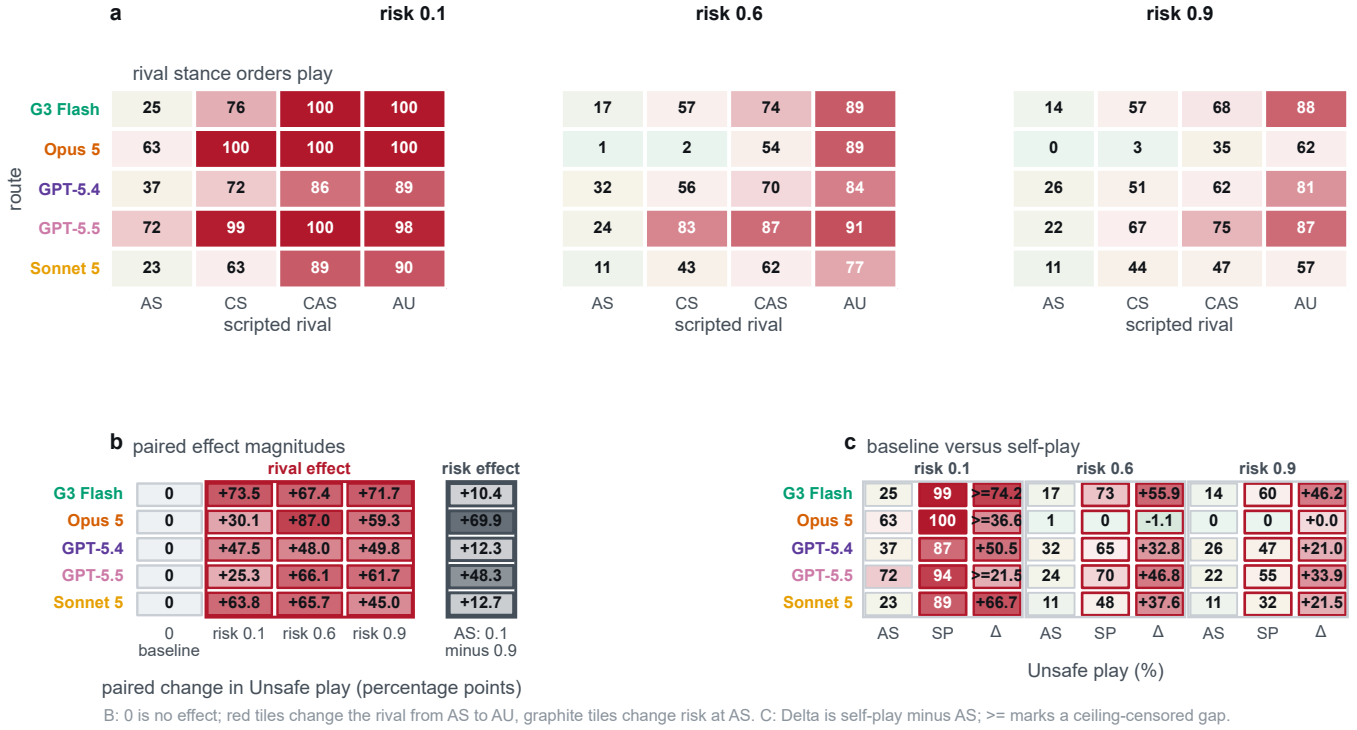


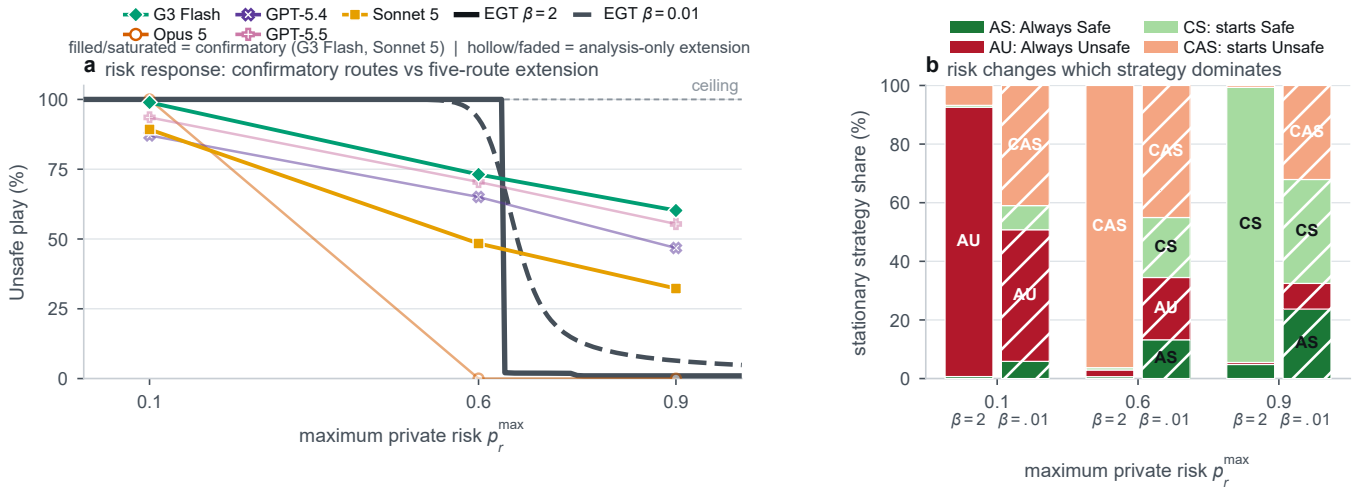
Figure 3: A shared rival response changes a route’s play. Panel (a) is a response surface: each tile is an admitted route’s Unsafe rate under one scripted rival, with columns ordered from Always Safe to Always Unsafe and facets for the three maximum-risk conditions. Panel (b) shows paired effect magnitudes as categorical tiles: the red block changes the rival from Always Safe to Always Unsafe, while the graphite tile changes stated risk with the rival held at Always Safe. Panel (c) shows the fixed-rival rate, self-play rate and their signed gap for each risk condition. These point estimates use the same matched blocks reported in the supplementary material; the corresponding paired 95% intervals are reported there as well. Within-route contrasts are estimands; cross-route magnitudes are descriptive under heterogeneous contracts.

The agents land nearer to the regime that describes the people than to the one usually quoted as the theory’s prediction, and that cell is not one we chose to fit: $\beta = 0.01$ and $\mu = 0.05$ are the values the source study reports as its own best fit to human play. An analysis-only extension fits the same existing sweep to all five admitted routes. For the four graded routes, the best well-mixed weak-selection cell has RMSEs of 5.1–15.4 percentage points, compared with 32.9–36.6 points at the reference cell; Claude Opus 5 is the qualitative exception because its observed profile is a near-step switch (100%, 0%, 0%), not a graded decline. The full five-route fit is in the supplement. The interpretation stays bounded. A stationary distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted decisions in self-play; the agreement in shape says the graded risk response is consistent with a near-neutral, high-exploration population, not that the routes are evolving. What the sweep establishes is narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

4.4 Frontier agents are behaviourally concentrated

Twenty people asked to run this race do not behave alike; twenty copies of a frontier route very nearly do. Human participants arrive at their average level of UNSAFE play through many different histories, while each admitted route reuses a small set of them, so a route can match the human mean without resembling the human population at all. We measure this on first-five-round trajectories. Each player contributes one exact paired trajectory, a ten-bit string read within its own risk condition; because the progress increments are fixed, that string fully specifies the first-five-round joint-action trajectory. The comparison covers all nine routes that faced the screen, admitted and refused alike, on the confirmatory baseline, 60 trajectories per route, against 340 complete human trajectories from the public de-identified dataset of Domingos and Han [14]. That single published study of a single game is the only human reference here, which bounds every human comparison below to that population and that mechanism.

Of 341 released participants, 340 provide complete first-five-round trajectories and 338 provide complete dynamic-model records. Among the complete trajectories, 168 source dyads contribute 336



Panel a shows the small-mutation limit; panel b shows the archived finite-mutation compositions at the reference and reported best-fit settings.

Figure 4: The evolutionary benchmark and the admitted route profiles in a common risk coordinate. (a) The direct confirmatory comparison is the separate two-route block for Gemini 3 Flash and Claude Sonnet 5; the other three admitted profiles are shown only as an analysis-only extension. The small-mutation limit is step-like at strong selection and graded near neutrality, while the graded profiles move gradually and Claude Opus 5 changes sharply between the two lowest tested levels. (b) The corresponding finite-mutation population composition changes its dominant strategy from Always Unsafe through the conditional strategy that starts Unsafe to the conditional strategy that starts Safe as risk rises. The hatched bars show the reported best-fit regime; the unhatched bars show the reference regime.

trajectories; four complete trajectories have no complete partner after de-identification and are not used in the dyad null. Before using this dataset as an external reference, we validated the reconstruction by refitting the source study’s published dynamic specification on its released data: 2,888 participant-round observations from 338 participants in 172 race-pair clusters (see the supplementary material).

Figure 5 counts distinct first-five-round trajectories in matched samples of twenty, against 20,000 human resamples of ten complete dyads at each risk. The human sample is larger, so matching prevents sample size from inflating its diversity, while dyad sampling preserves the unit of interaction. Eighteen of 27 route cells, including 13 of the 15 admitted cells, fall below every dyad draw. The participant-level sensitivity gives 21 of 27, including all 15 admitted cells; the primary dyad-null floors are 12, 15 and 13 distinct trajectories at risks 0.1, 0.6 and 0.9. The complementary participant-level table records effective counts of 18.9, 19.7 and 19.3 and mean pairwise distances of 0.47, 0.50 and 0.50. Panel b sorts the observed paths by frequency and panel c displays the modal five-round joint-action sequence. The same cells support two numerical checks reported in the supplement: Hill $q=1$ counts effective trajectories and mean pairwise Hamming distance measures separation of their ten action bits. Under that complementary Hill $q=1$ analysis, every admitted route sits entirely below the human interval at every risk level.

The concentration is strongest for Claude Opus 5, which produces a single trajectory, repeated by all 20 of its trajectories, at every risk level, while Claude Sonnet 5 uses two at risk 0.1. Concentration is also distinct from level: Gemini 3 Flash and GPT-5.5 finish within five points of each other at the highest risk and use a

similar number of distinct trajectories, 12 and 13, yet the Gemini trajectories sit half again as far apart from one another, 0.41 against 0.27 in mean pairwise distance.

Under the participant-level sensitivity, two checkpoints are the only exceptions, and the screen refuses both of them. GPT-5.4 mini uses 20, 17 and 17 distinct trajectories out of 20 at risks 0.1, 0.6 and 0.9, and GPT-5.4 nano 16, 19 and 16; both land inside the participant-level human null at every risk level, and no other route reaches it once under that sensitivity. On this measure they are not less diverse than people. Under the primary dyad null, GPT-5.4 and GPT-5.5 also meet the finite-draw floor at risk 0.9, so the cluster-aware headline is 13 of 15 admitted cells rather than all 15. Both refused routes are routes the quality-control screen of §3.3 refused, at 75.0% and 51.7% overall probe accuracy. That is why the screen is reported in the body: without it, these two rows would read as the finding that small models match human strategic diversity.

The same picture holds one level down, on the rates themselves. Raw player-level UNSAFE rates, plotted per risk condition in the supplementary material, spread across nearly the whole available range for humans at every risk level while each checkpoint occupies a narrow band of its own. Similarity between a checkpoint’s aggregate UNSAFE rate and the human mean therefore does not imply that it reproduces the human distribution. A two-dimensional embedding of the same space is in the supplementary material; we rest no conclusion on it, because recolouring the same coordinates by outcome separates them almost entirely by UNSAFE rate.

Populations that look alike in rate can still be following different observed rules. Exploratory classifier and feature analyses in the supplement suggest that possibility, but use a seven-checkpoint

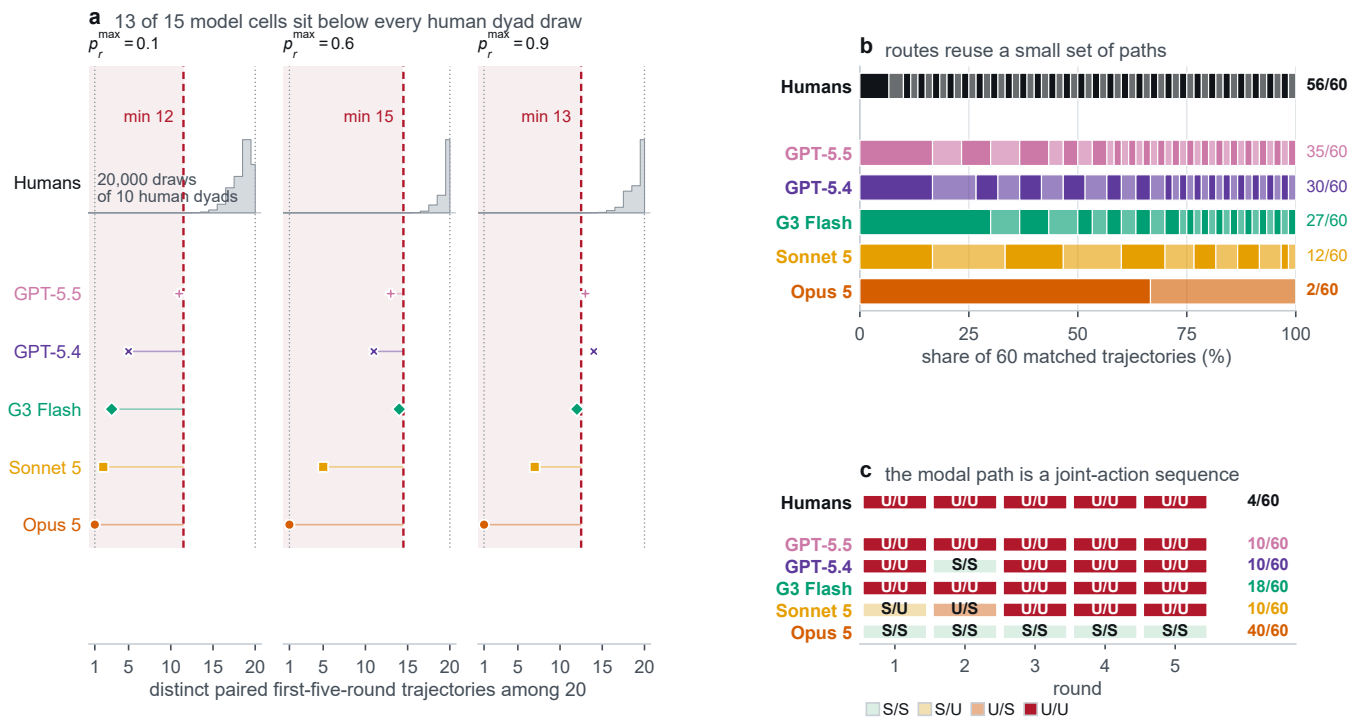


Figure 5: Human participants occupy a broader observed trajectory space than most admitted route-risk cells under a dyad-clustered null. Panel a compares distinct paired first-five-round trajectories with 20,000 draws of ten complete human dyads; 13 of 15 admitted cells fall below every dyad draw. The participant-level null is a sensitivity analysis and places all 15 below every draw. Panel b sorts the 60 matched trajectories by path frequency; panel c shows the modal pooled joint-action path. Each tile is the focal route’s action followed by its opponent’s.

roster containing neither GPT-5.4 nor GPT-5.5, so they support no claim about the admitted panel.

5 DISCUSSION AND CONCLUSION

These agents incorporate both the stated risk and the rival’s stance. UNSAFE play is non-increasing as the stated chance of catastrophe rises on every admitted route, while on the five routes where the opponent was replaced by scripted code, the rival’s stance changes behaviour substantially, with a magnitude that varies by route and can overlap the effect of changing risk. In this game, the five tested routes are therefore strongly context-sensitive: how the other side behaves can be more informative than the risk level alone.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where a route matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result.

Four limitations bound the claims directly. Nine commercial endpoints are not a sample from a population of models, and any two of them differ in far more than the one property a comparison names, so every contrast drawn between routes in this paper is descriptive and none of it separates a route’s behaviour from its provider, its scale or its training. The scripted-rival campaign covers five routes and four reduced strategies, which are four points in a space of

policies rather than a sample of opponents. The human comparison rests on one published study of one game, 340 complete trajectories, and we do not add a second human dataset, because any other dataset would change the game as well as the population. And the evolutionary comparison is a reconstruction audited against pinned EGTtools source rather than a bitwise reproduction of undisclosed author code, so it bounds the regime in which the benchmark is read rather than fitting the agents.

Four further limits are worth stating. Fixed-state replay and live play measure different things, one decision at a saved state against dependent decisions along a trajectory, so their effect sizes are not interchangeable. The representation-robustness design is crossed for two routes, enough to show the effect replicates but not to characterise endpoints in general. A matched group-size grid now exists on one route, but adding a company changes the stage payoff and the prompt length too, so no pure group-size effect is identifiable. Seed forwarding is provider-specific: one route’s SDK stripped the requested seed and another forwarded it unconfirmed. And this game models a private speed-versus-safety trade-off and nothing else, with no collective harm, regulation, communication or uncertainty about real capabilities.

REFERENCES

- [1] Gati V Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 337–371. <https://proceedings.mlr.press/v202/aher23a.html>
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (01 Jul 2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] Jacy Reese Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, Erik Brynjolfsson, James Evans, and Michael S. Bernstein. 2025. Position: LLM Social Simulations Are a Promising Research Method. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*, PMLR, Vancouver, Canada, 81005–81034. <https://proceedings.mlr.press/v267/anthis25a.html>
- [4] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023), 337–351. doi:10.1017/pan.2023.2
- [5] Stuart Armstrong, Nick Bostrom, and Carl Shulman. 2016. Racing to the precipice: a model of artificial intelligence development. *AI & SOCIETY* 31, 2 (01 May 2016), 201–206. doi:10.1007/s00146-015-0590-y
- [6] Amanda Askell, Miles Brundage, and Gillian K. Hadfield. 2019. The Role of Cooperation in Responsible AI Development. arXiv:1907.04534 [cs.CY] <https://arxiv.org/abs/1907.04534>
- [7] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis* 32, 4 (2024), 401–416. doi:10.1017/pan.2024.5
- [8] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. 2025. FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory. arXiv:2504.14325 [cs.AI] doi:10.48550/arXiv.2504.14325 Preprint, version 5 revised March 2026.
- [9] Yong Cao, Haijiang Liu, Arnab Arora, Isabelle Augenstein, Paul Röttger, and Daniel Hershcovich. 2025. Specializing Large Language Models to Simulate Survey Response Distributions for Global Populations. In *Proceedings of NAACL 2025*. Association for Computational Linguistics, Albuquerque, New Mexico, 3141–3154. doi:10.18653/v1/2025.naacl-long.162
- [10] Stephen Cave and Seán S. ÓhÉigeartaigh. 2018. An AI Race for Strategic Advantage: Rhetoric and Risks. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (New Orleans, LA, USA) (AIES '18). Association for Computing Machinery, New York, NY, USA, 36–40. doi:10.1145/3278721.3278780
- [11] Theodor Cîmpeanu, Francisco C. Santos, Luís Moniz Pereira, Tom Lenaerts, and The Anh Han. 2022. Artificial intelligence development races in heterogeneous settings. *Scientific Reports* 12, 1 (02 Feb 2022), 1723. doi:10.1038/s41598-022-05729-3
- [12] Julian Coda-Forno, Marcel Binz, Jane X. Wang, and Eric Schulz. 2024. CogBench: A Large Language Model Walks into a Psychology Lab. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*, PMLR, Vienna, Austria, 9076–9108. <https://proceedings.mlr.press/v235/coda-forno24a.html>
- [13] R. Maria del Río-Chanona, Marco Pangallo, and Cars Hommes. 2025. Can Generative AI Agents Behave Like Humans? Evidence from Laboratory Market Experiments. arXiv:2505.07457 [econ.GN] <https://arxiv.org/abs/2505.07457> Preprint.
- [14] Elías Fernández Domingos and The Anh Han. 2026. Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment. arXiv:2607.26034 [cs.AI] <https://arxiv.org/abs/2607.26034>
- [15] Salvatore Giorgi, Tingting Liu, Ankit Aich, Kelsey Jane Isman, Garrick Sherman, Zachary Fried, João Sedoc, Lyle Ungar, and Brenda Curtis. 2024. Modeling Human Subjectivity in LLMs Using Explicit and Implicit Human Factors in Personas. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 7174–7188. doi:10.18653/v1/2024.findings-emnlp.420
- [16] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. 2020. To regulate or not: A social dynamics analysis of an idealised ai race. *Journal of Artificial Intelligence Research* 69 (2020), 881–921. doi:10.1613/JAIR.1.12225
- [17] Nathan Herr, Fernando Acero, Roberta Raileanu, María Pérez-Ortiz, and Zhibin Li. 2024. Are Large Language Models Strategic Decision Makers? A Study of Performance and Bias in Two-Player Non-Zero-Sum Games. arXiv:2407.04467 [cs.AI] <https://arxiv.org/abs/2407.04467>
- [18] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2025. Understanding LLM Agent Behaviours via Game Theory: Strategy Recognition, Biases and Multi-Agent Dynamics. arXiv:2512.07462 [cs.MA] doi:10.48550/arXiv.2512.07462 Preprint.
- [19] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2026. Payoff Scaling Shapes Cooperation in LLM Agents across Languages. arXiv:2601.19082 [cs.AI] doi:10.48550/arXiv.2601.19082 Preprint, version 2 revised June 2026.
- [20] Paraskevas V. Lekeas and Giorgos Stamatoopoulos. 2026. What Suppresses Nash Equilibrium Play in Large Language Models? Mechanistic Evidence and Causal Control. arXiv:2604.27167 [cs.AI] <https://arxiv.org/abs/2604.27167>
- [21] Yuxuan Lu, Jing Huang, Yan Han, Bingsheng Yao, Sisong Bei, Yaochen Xie, Yisi Sang, Qi He, and Dakuo Wang. 2026. Can LLM Agents Simulate Multi-Turn Human Behavior? Evidence from Real Online Customer Behavior Data. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, San Diego, California, USA, 43961–43977. doi:10.18653/v1/2026.acl-long.2034
- [22] Saptarshi Pal, Abhishek Mallela, Christian Hilbe, Lenz Pracher, Chiyu Wei, Feng Fu, Santiago Schnell, and Martin A. Nowak. 2026. Strategies of Cooperation and Defection in Five Large Language Models. arXiv:2601.09849 <https://arxiv.org/abs/2601.09849>
- [23] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 [cs.AI] <https://arxiv.org/abs/2507.02618>
- [24] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, Mexico City, Mexico, 2006–2017. doi:10.18653/v1/2024.findings-naacl.130
- [25] Isaac Robinson and John Burden. 2025. Framing the Game: How Context Shapes LLM Decision-Making. arXiv:2503.04840 [cs.CL] <https://arxiv.org/abs/2503.04840>
- [26] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. arXiv:2401.03729 [cs.CL] <https://arxiv.org/abs/2401.03729>
- [27] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying About Prompt Formatting. In *International Conference on Learning Representations*. OpenReview.net, Vienna, Austria. https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html
- [28] Jia Wang, Ziyu Zhao, Tingjuntao Ni, and Zhongyu Wei. 2025. SocioBench: Modeling Human Behavior in Sociological Surveys with Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Suzhou, China, 26257–26289. doi:10.18653/v1/2025.emnlp-main.1335
- [29] Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling Selection Biases: Exploring Order and Token Sensitivity in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 5598–5621. doi:10.18653/v1/2024.findings-acl.333
- [30] Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Mexico City, Mexico, 1819–1862. doi:10.18653/v1/2024.naacl-long.102
- [31] Jiayi Yao, Cong Chen, and Baosen Zhang. 2026. Competition and Cooperation of LLM Agents in Games. arXiv:2604.00487 <https://arxiv.org/abs/2604.00487>
- [32] Kehan Zheng, Jinfeng Zhou, and Hongning Wang. 2025. Beyond Nash Equilibrium: Bounded Rationality of LLMs and Humans in Strategic Decision-Making. arXiv:2506.09390 <https://arxiv.org/abs/2506.09390>
- [33] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. When “A Helpful Assistant” Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 15126–15154. doi:10.18653/v1/2024.findings-emnlp.888